Lab 9, Batch G1

Dated: 12-11-2026

1. Write a C++ program to construct and implement a Canonical LR(1) parser for the following grammar.

   E → E OR T
   E → T
   T → T AND F
   T → F
   F → NOT F
   F → (E)
   F → id

Your program must:

1. Augment the grammar.
2. Generate all Canonical LR(1) item sets.
3. Implement CLOSURE(I, X) using FIRST information and LR(1) lookaheads.
4. Implement GOTO(I, X).
5. Display every LR(1) item with its lookahead.
6. Construct the ACTION/GOTO table.
7. Detect shift-reduce and reduce-reduce conflicts.
8. Determine whether the grammar is LR(1).
9. Implement the LR(1) parsing algorithm.
10. Display the complete parsing trace.

Test Inputs

id OR id AND id
NOT id OR id
(id OR id) AND id
NOT (id AND id)

Expected Results

id OR id AND id → ACCEPTED
NOT id OR id → ACCEPTED
(id OR id) AND id → ACCEPTED
NOT (id AND id) → ACCEPTED

Test Invalid

id OR AND id

NOT OR id
(id OR id
id AND (id OR)

Expected:

id OR AND id → REJECTED
NOT OR id → REJECTED
(id OR id → REJECTED
id AND (id OR) → REJECTED

Instructions

Due date: 12.11.2026, 11:30 A.M for Batch G1 (30 minutes extra is given for the documentation)

Cut-off date: 13.11.2026, 11:50 P.M. (Late submission deadline)

Note:

1. If the submission is after the due date/time, marks will be deducted for every 6 hours till the cut-off date. After the Cut-off date, No evolution and no marks
2. If the output is not shown for any of the above components, the mark cannot be given even if it is included in your documentation.